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CS499

Professional Self-Assessment

Completing the Bachelor of Science in Computer Science program at Southern New Hampshire University has significantly strengthened my technical knowledge, problem-solving abilities, and professional confidence. Throughout the program, I have developed skills in software engineering, full-stack development, database management, computational graphics, artificial intelligence, cybersecurity, and systems design. The process of creating my ePortfolio has allowed me to reflect on my growth as a computer science professional while demonstrating the knowledge and skills I have acquired through both coursework and practical application. This portfolio showcases my ability to design, develop, test, and improve software solutions while preparing me to pursue opportunities within software development, cybersecurity, network engineering, and related technology fields.

One of the most valuable aspects of the Computer Science program was learning how to collaborate and communicate effectively with various stakeholders. While many assignments were completed individually, coursework frequently required me to analyze user requirements, consider business needs, and design solutions that balanced technical functionality with usability. Projects such as the Travlr Getaways full-stack application required me to think from both the developer and end-user perspectives. Through project documentation, presentations, code reviews, and written analyses, I learned how to communicate complex technical concepts in a way that is understandable to both technical and non-technical audiences. These experiences strengthened my ability to gather requirements, provide status updates, document solutions, and explain technical decisions to stakeholders.

Throughout the program, I gained extensive experience applying data structures and algorithms to improve software performance and scalability. Courses focused on software development and algorithm analysis introduced concepts such as searching, sorting, collection management, recursion, and efficiency optimization. Within my Travlr Getaways enhancement project, I implemented pagination, search functionality, and sorting algorithms to improve database retrieval performance and reduce unnecessary processing. These enhancements demonstrated how thoughtful algorithm design can significantly improve application scalability and user experience. Learning to evaluate algorithm efficiency and select appropriate data structures has strengthened my ability to design solutions that remain effective as applications and datasets grow.

The program also provided substantial experience in software engineering principles and practices. Through projects involving Java, JavaScript, Angular, Node.js, Express, and other technologies, I developed a deeper understanding of software architecture, modular design, object-oriented programming, testing, debugging, and maintainability. The Travlr Getaways application serves as a strong example of my software engineering abilities because it combines frontend development, backend API development, authentication, database integration, and user interface design into a complete full-stack solution. The software engineering enhancement further demonstrated my ability to evaluate existing codebases and implement improvements that increase maintainability, scalability, and overall software quality.

Database design and management became another important area of growth throughout the program. Working with MongoDB and Mongoose allowed me to gain practical experience designing schemas, implementing validation rules, managing collections, and optimizing database performance. Through my database enhancement project, I improved data integrity by implementing schema validation, indexing strategies, duplicate record prevention, and enhanced error handling. These enhancements demonstrated the importance of maintaining reliable and secure data storage systems while improving application performance. Understanding how software interacts with databases has strengthened my ability to design robust data-driven applications.

Security has also become a critical component of my technical skillset. Throughout the program, I learned how security principles apply to software development, authentication systems, network communication, and data management. In the Travlr Getaways project, I implemented JSON Web Token (JWT) authentication to protect administrative functions and restrict access to authorized users. Additional coursework reinforced concepts such as secure coding practices, access control, input validation, and risk mitigation. These experiences have increased my awareness of cybersecurity concerns and strengthened my ability to develop systems that prioritize confidentiality, integrity, and availability.

The artifacts included within this ePortfolio collectively demonstrate the breadth of my computer science knowledge and skills. The Software Design and Engineering artifact highlights my ability to develop and enhance full-stack applications using modern development frameworks and software engineering practices. The Algorithms and Data Structures artifact demonstrates my ability to evaluate performance, implement scalable retrieval methods, and improve application efficiency through algorithmic enhancements. The Database artifact showcases my understanding of schema design, validation, indexing, and data integrity within a MongoDB environment. Together, these artifacts illustrate my ability to analyze existing systems, identify opportunities for improvement, and implement meaningful technical solutions that improve functionality, performance, security, and maintainability.

Overall, the Computer Science program and the development of this ePortfolio have prepared me to enter the technology field with confidence. The combination of technical coursework, hands-on projects, and reflective analysis has strengthened my problem-solving abilities and reinforced my commitment to continuous learning. As technology continues to evolve, I am prepared to apply the knowledge and skills gained throughout this program to contribute effectively within software development, cybersecurity, network engineering, and other technology-focused roles. This portfolio represents not only the work I have completed during my academic journey but also the foundation upon which I will continue building my professional career.